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VLSI Design
Second Computer Assignment
Delay Optimization and Logical Effort

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2025-2026

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1 Manual and Numerical Delay Calculations

In this section, our goal is to design the gate dimensions to minimize the delay in the specified path from input A to output out. For this purpose, we use the Logical Effort method.

1.1 Definition of the Analyzed Path

The path intended for optimization is highlighted in red in the following figure. This path starts from input A and ends at the out node.

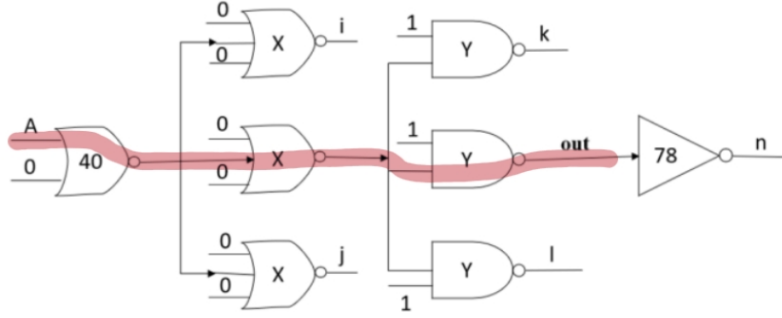


Figure 1: Critical path from input A to output out

The gates located in this path, whose dimensions we will design to minimize delay, are as follows in order:

1. Two-input NOR gate
2. Three-input NOR gate (X)
3. Two-input NAND gate (Y)

Therefore, our path consists of 3 stages. The final inverter gate is considered as the output load for the last stage NAND gate.

1.2 Parameter Calculation and Sizing

The basis of the logical effort method is comparing all gates with a basic Inverter. Assuming $\mu_n \approx 2\mu_p$, the dimensions of the reference inverter are determined as follows:

- NMOS transistor width: $W_n = 1$
- PMOS transistor width: $W_p = 2$

Based on this, the input capacitance of the reference inverter, which is used as the basis for all subsequent calculations, is equal to:

$$C_{in,inv} = W_p + W_n = 2 + 1 = 3 \text{ units}$$

The delay of each stage in the circuit is modeled by four main parameters:

$$g = \frac{C_{\text{in,gate}}}{C_{\text{in,inv}}}$$

$$p = \frac{C_{\text{out,diff,gate}}}{C_{\text{in,inv}}}$$

$$b = \frac{C_{\text{on-path}} + C_{\text{off-path}}}{C_{\text{on-path}}} = \frac{C_{\text{total}}}{C_{\text{on-path}}}$$

$$h = \frac{C_{\text{load}}}{C_{\text{in,gate}}}$$

1.2.1 Two-Input NOR Gate

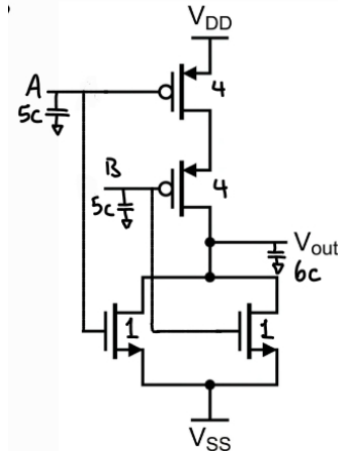


Figure 2: Schematic view of the two-input NOR gate

Sizing:

- Pull-up: Consists of two series PMOS transistors. To have an equivalent resistance of a PMOS with a width of 2, each must have a width of $2 \times 2 = 4$.
- Pull-down: Consists of two parallel NMOS transistors. In the worst case, only one of them is on, so to have an equivalent resistance of an NMOS with a width of 1, its dimension will be 1.

Parameter Calculation:

$$\begin{aligned}g_1 &= g_{\text{NOR2}} = \frac{C_{\text{in,NOR2}}}{C_{\text{in,INV}}} = \frac{5}{3} \\p_1 &= p_{\text{NOR2}} = \frac{C_{\text{out,NOR2}}}{C_{\text{out,INV}}} = \frac{6}{3} = 2 \\b_1 &= \frac{X + X + X}{X} = \frac{3X}{X} = 3 \\h_1 &= \frac{C_{\text{load,on-path}}}{C_{\text{in,gate}}} = \frac{3X}{40}\end{aligned}$$

1.2.2 Three-Input NOR Gate (X)

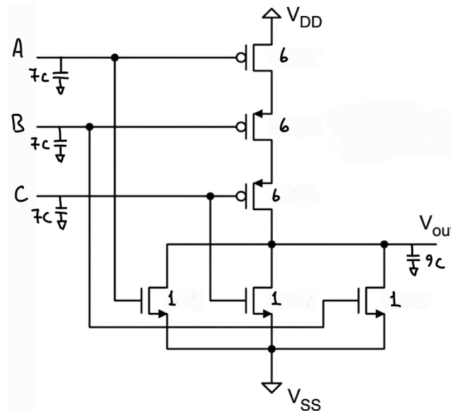


Figure 3: Schematic view of the three-input NOR gate (X)

Sizing:

- Pull-up: Three series PMOS transistors require a width of $3 \times 2 = 6$ for each.
- Pull-down: Three parallel NMOS transistors, similar to before, require a width of 1 for each.

Parameter Calculation:

$$\begin{aligned}g_2 &= g_{\text{NOR3}} = \frac{C_{\text{in,NOR3}}}{C_{\text{in,INV}}} = \frac{7}{3} \\p_2 &= p_{\text{NOR3}} = \frac{C_{\text{out,NOR3}}}{C_{\text{out,INV}}} = \frac{9}{3} = 3 \\b_2 &= \frac{Y + Y + Y}{Y} = \frac{3Y}{Y} = 3 \\h_2 &= \frac{C_{\text{load,on-path}}}{C_{\text{in,gate}}} = \frac{3Y}{X}\end{aligned}$$

1.2.3 Two-Input NAND Gate (Y)

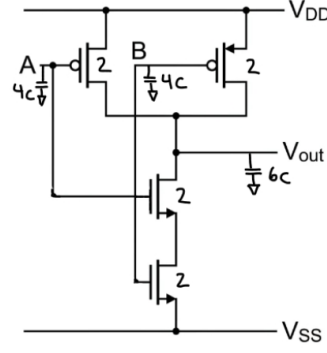


Figure 4: Schematic view of the two-input NAND gate (Y)

Sizing:

- Pull-up: Two parallel PMOS transistors; in the worst case, one of them is on. To have an equivalent resistance of a PMOS with a width of 2, its dimension will be 2.
- Pull-down: Two series NMOS transistors; to have an equivalent resistance of an NMOS with a width of 1, each must have a width of $2 \times 1 = 2$.

Parameter Calculation:

$$g_3 = g_{\text{NAND2}} = \frac{C_{\text{in,NAND2}}}{C_{\text{in,INV}}} = \frac{4}{3}$$

$$p_3 = p_{\text{NAND2}} = \frac{C_{\text{out,NAND2}}}{C_{\text{out,INV}}} = \frac{6}{3} = 2$$

$$b_3 = 1$$

$$h_3 = \frac{C_{\text{load,on-path}}}{C_{\text{in,gate}}} = \frac{78}{Y}$$

1.3 Path Effort Analysis and Calculation

After calculating the basic parameters of each gate (g, p, b, h), the next step is calculating the total path effort, denoted by F . This parameter is defined for the entire path from input A to the out node and is obtained from the following main equation:

$$F = G \cdot B \cdot H$$

1.3.1 Calculating Path Logical Effort - G

This parameter is obtained by multiplying the g of each individual gate in the path:

$$G = \prod_{i=1}^N g_i$$

where N is the number of stages or gates in the path (here $N = 3$). We have three gates, NOR2, NOR3, and NAND2, in the path:

$$G = g_{\text{NOR2}} \times g_{\text{NOR3}} \times g_{\text{NAND2}}$$

By substituting the values calculated in the previous section:

$$G = \left(\frac{5}{3}\right) \times \left(\frac{7}{3}\right) \times \left(\frac{4}{3}\right) = \frac{5 \times 7 \times 4}{3 \times 3 \times 3} = \frac{140}{27}$$

1.3.2 Calculating Path Branching Effort - B

This parameter is obtained from the product of b of each stage:

$$B = \prod_{i=1}^N b_i$$

Calculation for the desired path:

$$B = b_1 \times b_2 \times b_3$$

By substituting the values:

$$B = 3 \times 3 \times 1 = 9$$

1.3.3 Calculating Path Electrical Effort - H

The electrical effort of the entire path is generally defined as the ratio of the final load capacitance at the end of the path to the input capacitance at the beginning of the path:

$$H = \frac{C_{\text{out-path}}}{C_{\text{in-path}}}$$

where:

- $C_{\text{out-path}}$: is the load capacitance at the end of the path (here, the input capacitance of the final inverter, which is 78).
- $C_{\text{in-path}}$: is the input capacitance of the first gate in the path (here, the input capacitance of the NOR2 gate, which is 40).

Calculation for the desired path:

$$H = \frac{78}{40}$$

1.3.4 Final Calculation of Total Path Effort (F)

Now, having the values of G , B , and H , we can calculate the total path effort F :

$$\begin{aligned} F &= G \cdot B \cdot H \\ F &= \left(\frac{140}{27}\right) \times 9 \times \left(\frac{78}{40}\right) \\ F &= \left(\frac{140}{3}\right) \times \left(\frac{78}{40}\right) = 91 \\ F &= 91 \end{aligned}$$

1.4 Delay Optimization and Optimal Sizing

The fundamental principle in optimizing the delay of a logical path is that the actual effort of each stage (Stage Effort) must be equal. This optimal value, which we call $\hat{f}$, is obtained from the N -th root of F :

$$\hat{f} = F^{1/N}$$

where $F = 91$ and $N = 3$. The calculations are as follows:

$$\hat{f} = (91)^{1/3} \approx 4.498 \approx 4.5$$

Now, to find the optimal dimensions, we set the actual effort of each stage (including $f_i = g_i \cdot h_i \cdot b_i$) equal to the target value $\hat{f}$ and calculate the unknowns one by one:

a) Calculating X through the first stage

By setting $f_1 = \hat{f}$:

$$\begin{aligned} f_1 &= g_1 \cdot h_1 = \hat{f} \\ \left(\frac{5}{3}\right) \cdot \left(\frac{3X}{40}\right) &= 4.5 \\ 5 \cdot \left(\frac{X}{40}\right) &= 4.5 \\ \frac{X}{8} &= 4.5 \\ X = 8 \times 4.5 &\implies X \approx 36 \end{aligned}$$

b) Calculating Y through the third stage

By setting $f_3 = \hat{f}$:

$$\begin{aligned} f_3 &= g_3 \cdot h_3 = \hat{f} \\ \left(\frac{4}{3}\right) \cdot \left(\frac{78}{Y}\right) &= 4.5 \\ \frac{104}{Y} &= 4.5 \\ Y = \frac{104}{4.5} &\implies Y \approx 23.11 \approx 23 \end{aligned}$$

1.5 Minimum Delay Calculation

The general formula for the delay of a path is as follows:

$$D = \sum_{i=1}^N (f_i + p_i)$$

Since the goal of optimization was to equalize the effort in all stages ($f_i = f_{\text{optimal}}$), the formula takes the following simpler form:

$$D_{\min} = N \cdot f_{\text{optimal}} + P_{\text{path}}$$

where $P_{\text{path}} = \sum p_i$ is the sum of parasitic delays of the entire path. According to the calculated values:

$$P_{\text{path}} = p_{\text{NOR2}} + p_{\text{NOR3}} + p_{\text{NAND2}} = 2 + 3 + 2 = 7$$

By substituting the above values into the formula, we have:

$$D_{\min} = (N \cdot f_{\text{optimal}}) + P_{\text{path}}$$

$$D_{\min} = (3 \times 4.5) + 7$$

$$D_{\min} = 13.5 + 7 = 20.5$$

2 Circuit Design in Hspice Software

2.1 Definition of Base Transistor Dimensions and Calculation of the Size Parameter for Each Gate in the Critical Path

The transistor dimensions for the standard gates used in this project were determined in the first section of the report. The results of those calculations, which form the basis for designing the subcircuits in Hspice, are summarized in the following table for reference:

Table 1: Base transistor dimensions for standard gates

Gate Type	W_n	W_p
INV	1	2
NAND2	2	2
NOR2	1	4
NOR3	1	6

The formula for calculating the size for each gate is defined as follows:

$$\text{size} = \frac{\text{Optimal input capacitance (calculated)}}{\text{Input capacitance of the base gate}}$$

Using the optimal values $X \approx 36$ and $Y \approx 23$ obtained in the delay analysis section, we calculate the size value for each of the path gates:

1. First gate (NOR2):

- Required input capacitance: $C_{\text{in},1} = 40$
- Input capacitance of the base NOR2 gate: $C_{\text{base,NOR2}} = 5$

$$\text{size}_1 = \frac{40}{5} = 8$$

2. Second gate (NOR3 - gate X):

- Required input capacitance: $C_{\text{in},2} = X \approx 36$
- Input capacitance of the base NOR3 gate: $C_{\text{base,NOR3}} = 7$

$$\text{size}_2 = \frac{36}{7} \approx 5.14$$

3. Third gate (NAND2 - gate Y):

- Required input capacitance: $C_{\text{in},3} = Y \approx 23$
- Input capacitance of the base NAND2 gate: $C_{\text{base,NAND2}} = 4$

$$\text{size}_3 = \frac{23}{4} = 5.75$$

4. Output load (Inverter): To ensure correct loading, we also calculate the size of the final load inverter:

- Required input capacitance: $C_{\text{load}} = 78$
- Input capacitance of the base Inverter: $C_{\text{base,INV}} = 3$

$$\text{size}_{\text{load}} = \frac{78}{3} = 26$$

These base dimensions were implemented as parametric subcircuits in the gates.sp file. A parameter named size with a default value of 1 was defined for each gate. This parameter acts as a coefficient for the base dimensions (W_n and W_p). Thus, by calling each gate in the main circuit, its dimensions can be easily adjusted to the value obtained from the theoretical analysis by specifying the size value.

2.2 Analysis of Simulation Results and Circuit Performance Verification

The purpose of this section is to fully simulate the combinational circuit described in the project requirements by applying the dimensional coefficients calculated in the first stage.

For this purpose, the delay.sp netlist file was created, in which the instantiation of gates with optimal dimensions, the definition of the input pulse source, and .measure commands are implemented. To stimulate the circuit, a pulse signal with a period of 4 nanoseconds and rise/fall times of 0.01 nanoseconds was applied to input A. The simulation was run for 8 nanoseconds to observe two full periods of signal behavior in the circuit.

The following figure shows the voltage waveforms at the main nodes of the critical path (input A, first gate output (node_1), second gate output (node_X_mid), third gate output (out), and final output (n)):

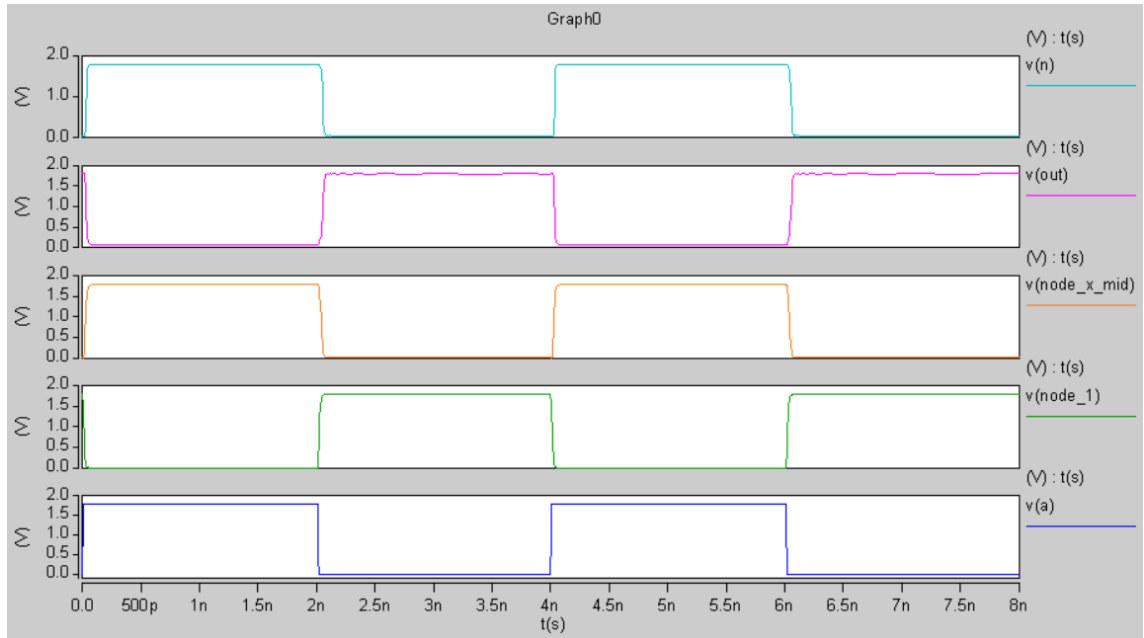


Figure 5: Voltage waveforms of main nodes in the critical path

2.2.1 Verification and Validation of Circuit Stages Performance

By examining the above waveforms, the correct operation of each circuit stage can be individually verified:

1. Verifying NOR2 gate performance:

- **Input:** A
- **Output:** node_1
- **Analysis:** Given that the other input of the NOR2 gate is connected to ground, this gate acts as an inverter. As seen in Figure 5, the node_1 signal shows exactly the inverted state of the input signal A with a slight propagation delay. This confirms the correct operation of the first gate.

2. Verifying NOR3 gate performance:

- **Input:** node_1
- **Output:** node_X_mid
- **Analysis:** This gate also acts as an inverter for its input signal (node_1) because its other two inputs are connected to ground. In the waveforms, it is clearly evident that the node_X_mid signal is the inverse of the node_1 signal and is consequently in-phase with the main signal A. The correct operation of this stage can also be verified by checking the branched outputs i and j, which should have behavior similar to node_X_mid.

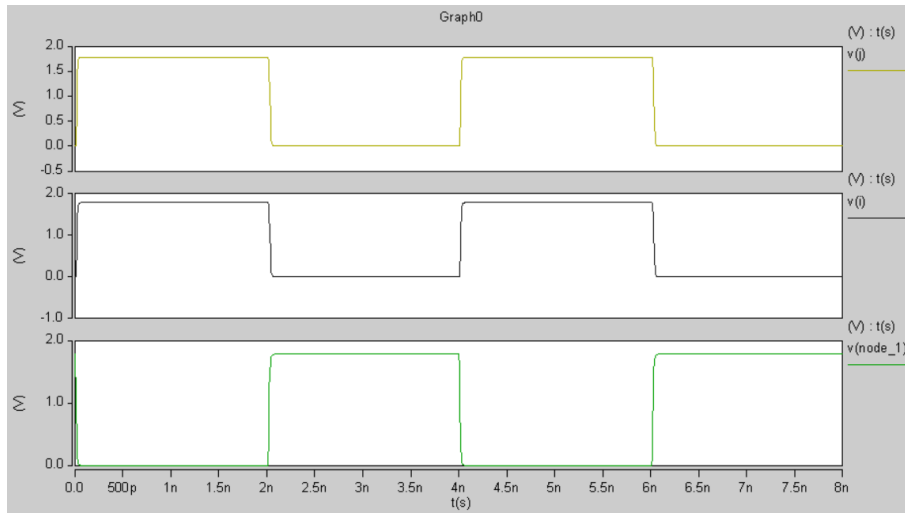


Figure 6: Voltage waveforms of nodes i and j

3. Verifying NAND2 gate performance:

- **Input:** node_X_mid
- **Output:** out
- **Analysis:** The other input of this gate is connected to VDD so that this gate also acts as an inverter. As expected, the out waveform shows the inverted state of node_X_mid. This behavior confirms the correct operation of the NAND2 gate. Examining outputs k and l will also show the same result.

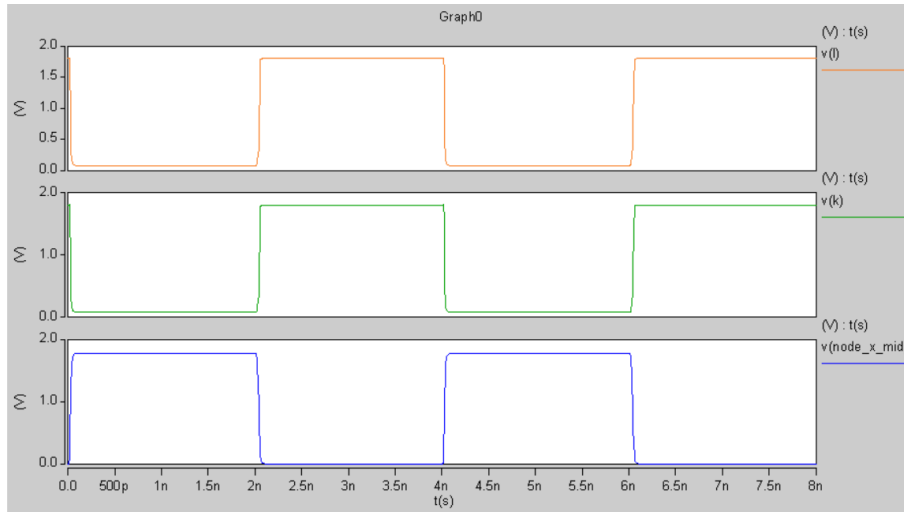


Figure 7: Voltage waveforms of nodes k and l

4. Verifying Inverter performance:

- **Input:** out
- **Output:** n
- **Analysis:** The final inverter is responsible for inverting the out signal. As seen in the waveforms, signal n is exactly the inverse of the out signal, which confirms the correct operation of the last stage.

2.2.2 Measurement Results of Timing Parameters and Comparison with Theory

Using the .measure commands in the Hspice simulation, key timing parameters of the circuit including propagation delay (t_{pHL} and t_{pLH}), average propagation delay ($t_{propagation}$), and rise and fall times (t_{rise} and t_{fall}) were extracted with high precision. The exact measured results are summarized in the following table:

Table 2: Extracted timing parameters from Hspice simulation

Parameter	Measured Value
t_{pHL}	28.00 ps
t_{pLH}	38.41 ps
$t_{propagation}$	33.21 ps
t_{rise}	39.09 ps
t_{fall}	22.80 ps

2.3 Speed-Power Trade-off Analysis Using Gate Sizing Technique

In this section, the goal was to practically investigate the effect of changing the dimensions of one of the key path gates on the overall circuit performance. Specifically, the NOR3 gate was selected for this analysis to obtain a trade-off between propagation delay and average power consumption, and to determine the optimal operating point for the circuit. For this purpose, while keeping the dimensions of other gates constant based on the optimal values calculated in the previous section, the size of the NOR3 gate was swept over a range around the theoretical optimum (2.5 to 7.5) using the .ALTER command in Hspice. The code for this simulation is implemented in the power.sp file. At each step, key parameters including total path propagation delay (t_p), rise and fall times at the final output (t_{rise} and t_{fall}), and average power consumption (P_{avg}) were measured. Complete results are provided in Table 3.

Table 3: Speed-power trade-off analysis results with varying NOR3 gate size

Gate Size (Size)	Propagation Delay (t_p) (ps)	Rise Time (t_{rise}) (ps)	Fall Time (t_{fall}) (ps)	Average Power (P_{avg}) (μW)
2.5	39.38	19.22	28.86	550.1
3.0	37.28	17.78	27.28	552.7
3.5	36.37	17.29	26.48	557.0
4.0	35.51	16.36	25.04	561.6
4.5	35.11	16.09	25.00	566.6
5.14	35.14	15.47	24.76	574.1
5.5	35.14	15.15	24.50	578.3
6.0	35.20	14.88	24.15	584.2
6.5	35.31	14.86	23.99	590.4
7.0	35.46	14.67	23.96	596.7
7.5	35.72	14.42	24.07	603.2

As expected, with an increase in gate size, the average power consumption increases uniformly and almost linearly. Unlike power, the delay behavior is non-uniform. As the gate size increases, the delay initially decreases and reaches a minimum at Size = 4.5 (with a delay of 35.11 ps). After this point, making the gate larger causes the delay to increase again. This analysis practically showed that to optimize the speed of a logical path, indiscriminately enlarging gates is not a suitable solution, and there is an optimal size to achieve the minimum delay. Furthermore, it confirmed a fundamental trade-off between speed and power; the optimal point for delay (Size = 4.5) is not necessarily the most power-efficient option.

2.4 Measurement of Power Consumption in Constant Input State

Following the project procedure to investigate power consumption in the steady state, the circuit input (A) was set to a constant DC value equal to 0. The code for this simulation is implemented in a file named power_static.sp. After running the simulation and sweeping the size of the NOR3 gate across 10 different values, the total power consumption of the circuit for each size was obtained as shown in Table 4.

Table 4: Total power consumption results with constant input

NOR3 Gate Size (Size)	Measured Power (μW)
2.5	392.4
3.0	394.1
3.5	395.7
4.0	397.3
4.5	398.9
5.5	402.2
6.0	403.8
6.5	405.5
7.0	407.1
7.5	408.7

To analyze these results, one must consider the nature of power consumption in CMOS circuits. The total power (P_{Total}) consists of two main components:

- **Dynamic Power (P_{Dynamic}):** Which is consumed when gates switch.
- **Static Power (P_{Static}):** Which is due to leakage currents in the steady state.

Since the input is constant in this simulation and no switching occurs in the internal nodes of the circuit, the dynamic power component is completely eliminated ($P_{\text{Dynamic}} \approx 0$) and approaches zero. Therefore, the measured values in the table above, although reported as total power (avg_power) by the simulator, actually represent **pure static power** (leakage power) of the circuit.

As seen in the results, with the increase in the size of the NOR3 gate, static power also increases uniformly. It should be noted that for the initial optimal size (size = 5.14), this power was measured at $401.0 \mu W$.

In this section, the results obtained from the speed-power trade-off analysis (Section C, with PULSE input) and the static power measurement results (with constant DC input) are compared with each other to determine the contribution of each power component. To better display the differences and the trend of changes, a bar chart is used. The required power values for drawing this chart were extracted manually from the .lis output files of both simulations. Then, the chart plotting was implemented using the Python programming language in the plot.py file.

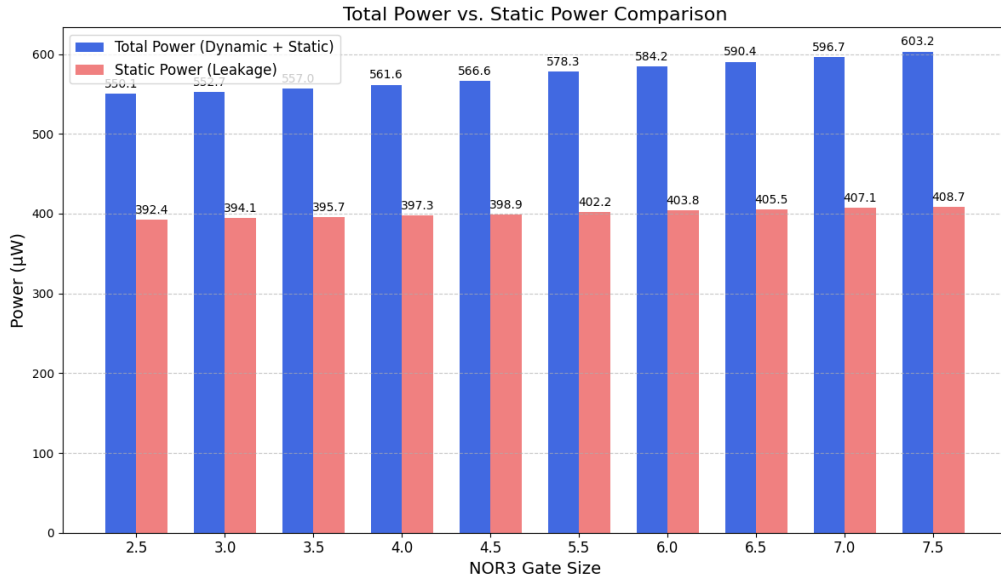


Figure 8: Comparative chart of Total power and Static power versus NOR3 gate size

The chart clearly shows that as the size of the NOR3 gate increases, both total power (blue) and static power (red) increase harmoniously. The most important conclusion derived from this comparison is the very high share of static power, which constitutes about 70% of the total power consumption of the circuit. This indicates that in modern technologies like 32nm, leakage power is no longer a marginal factor but has become the main challenge in designing low-power circuits. The difference between total and static power values represents dynamic power, which also grows with larger gates due to increased capacitance. The performance speed improvement achieved by enlarging the gates comes directly at the cost of a significant increase in power consumption, the bulk of which is due to leakage power.

Analysis of Dynamic Power Consumption Variations

In this section, we separate the dynamic component of power consumption from the total power and analyze its variation trend based on theoretical foundations. Dynamic power is calculated from the relationship $P_{\text{Dynamic}} = P_{\text{Total}} - P_{\text{Static}}$. The values extracted from the simulations and the calculated dynamic power are provided in Table 5.

Table 5: Total, static, and dynamic power results versus NOR3 gate size

Gate Size (Size)	Total Power (μW)	Static Power (μW)	Dynamic Power (μW)
2.5	550.1	392.4	157.7
3.0	563.8	394.1	169.7
3.5	579.0	396.1	182.9
4.0	595.0	398.3	196.7
4.5	611.8	400.7	211.1
5.14	629.4	401.0	228.4
5.5	647.6	403.5	244.1
6.0	666.3	406.4	259.9
6.5	685.7	409.5	276.2
7.0	705.5	412.8	292.7
7.5	725.9	416.2	309.7

As seen in the last column of the table, with the increase in the size of the NOR3 gate, the dynamic power value continuously increases. This result is perfectly consistent with the theoretical dynamic power relationship $P_{\text{Dynamic}} = \alpha C V_{DD}^2 f$.

In this simulation, the activity factor (α), supply voltage (V_{DD}), and frequency (f) are kept constant for all gate sizes. The only parameter that changes is capacitance (C). Increasing the gate size means increasing the width of its transistors, which directly leads to an increase in the internal and output capacitances of that gate.

Since dynamic power has a direct relationship with capacitance ($P_{\text{Dynamic}} \propto C$), increasing the gate size predictably leads to an increase in dynamic power consumption. The numerical results obtained in the table practically confirm this theoretical relationship.

3 Minimizing Total Circuit Delay

In the previous sections, we calculated the Path Effort for the circuit, which resulted in $F = 91$. Now we calculate the optimal value of $\hat{N}$:

$$\hat{N} = \log_{3.59} F$$

$$\hat{N} = \log_{3.59}(91) = \frac{\ln(91)}{\ln(3.59)} = \frac{4.51}{1.278} \approx 3.52$$

The optimal number for stages is $\hat{N} \approx 3.52$. Our current path has $N = 3$ stages. Since the optimal number (3.52) is greater than 3, we might be able to reduce the delay by adding one or more inverters (buffers) to the end of the path.

3.1 Calculations for the 4-Stage Path (N=4)

In this case, we add an inverter to the end of the main path. The new path:

NOR2 → NOR3 → NAND2 → INV

- **Path Logical Effort (G):**

$$G_{N=4} = g_{\text{NOR2}} \times g_{\text{NOR3}} \times g_{\text{NAND2}} \times g_{\text{INV}}$$

$$G_{N=4} = (5/3) \times (7/3) \times (4/3) \times 1 = 140/27$$

(The value of G does not change)

- **Path Branching Effort (B):**

$$B = 3 \times 3 \times 1 \times 1 = 9$$

- **Path Electrical Effort (H):**

$$H = C_{\text{out}}/C_{\text{in}} = 78/40 = 1.95$$

(Input and output load of the path remains constant)

- **Total Path Effort (F):**

$$F = G \times B \times H = (140/27) \times 9 \times 1.95 = 91$$

(The value of F remains constant)

- **Path Parasitic Delay (P):** By adding one inverter, its parasitic delay is added ($p_{\text{inv}} = 1$).

$$P_{N=4} = P_{N=3} + p_{\text{inv}} = 7 + 1 = 8$$

- **Optimal Effort of Each Stage ($\hat{f}$):**

$$\hat{f} = F^{1/N} = (91)^{1/4} \approx 3.088 \approx 3.1$$

- **Theoretical Minimum Delay (D_{min}):**

$$D_{\text{min},N=4} = N \cdot \hat{f} + P = (4 \times 3.1) + 8 = 12.4 + 8 = 20.4$$

This value (20.4) is less than the 3-stage delay (20.5), but there is a very important point: by adding only one inverter, the overall logic of the circuit is inverted. To maintain the correct operation of the circuit, inverters must be added in pairs. Therefore, this state is logically invalid, and we must proceed to the 5-stage case.

3.2 Calculations for the 5-Stage Path (N=5)

In this case, we add two inverters to the end of the main path to maintain the circuit logic. The new path: NOR2 → NOR3 → NAND2 → INV → INV

- **Path Logical Effort (G):**

$$G_{N=5} = g_{\text{NOR2}} \times g_{\text{NOR3}} \times g_{\text{NAND2}} \times g_{\text{INV}} \times g_{\text{INV}}$$

$$G_{N=5} = (5/3) \times (7/3) \times (4/3) \times 1 \times 1 = 140/27$$

(The value of G does not change)

- **Path Branching Effort (B):**

$$B = 3 \times 3 \times 1 \times 1 \times 1 = 9$$

- **Path Electrical Effort (H):**

$$H = C_{\text{out}}/C_{\text{in}} = 78/40 = 1.95$$

(Input and output load of the path remains constant)

- **Total Path Effort (F):**

$$F = G \times B \times H = (140/27) \times 9 \times 1.95 = 91$$

(The value of F remains constant)

- **Parasitic Delay (P):** The parasitic delay of two inverters is added to the previous sum.

$$P_{N=5} = P_{N=3} + p_{\text{inv}} + p_{\text{inv}} = 7 + 1 + 1 = 9$$

- **Optimal Effort of Each Stage ($\hat{f}$):**

$$\hat{f} = F^{1/N} = (91)^{1/5} \approx 2.46$$

- **Theoretical Minimum Delay (D_{min}):**

$$D_{\text{min},N=5} = N \cdot \hat{f} + P = (5 \times 2.46) + 9 = 12.3 + 9 = 21.3$$

To obtain the optimal dimensions, we set the effort of each stage equal to the optimal value ($\hat{f} = 2.46$). Our unknowns are the dimensions of the four gates in the path. We represent these dimensions with the letters X_{new} (NOR3), Y_{new} (NAND2), Z (INV1), and W (INV2). In the following calculations, these letters represent the input capacitance of the corresponding gate. Finally, we will convert these capacitances to the final size.

a) Calculating NOR3 input capacitance (Unknown X_{new})

By setting $f_1 = \hat{f}$:

$$\begin{aligned}
 f_1 &= g_1 \cdot h_1 = \hat{f} \\
 \left(\frac{5}{3}\right) \cdot \left(\frac{3 \cdot X_{new}}{40}\right) &= 2.46 \\
 \frac{5 \cdot X_{new}}{40} &= 2.46 \\
 \frac{X_{new}}{8} &= 2.46 \\
 X_{new} &= 8 \times 2.46 \implies X_{new} \approx 19.68
 \end{aligned}$$

b) Calculating NAND2 input capacitance (Unknown Y_{new})

By setting $f_2 = \hat{f}$:

$$\begin{aligned}
 f_2 &= g_2 \cdot h_2 = \hat{f} \\
 \left(\frac{7}{3}\right) \cdot \left(\frac{3 \cdot Y_{new}}{X_{new}}\right) &= 2.46 \\
 \frac{7 \cdot Y_{new}}{19.68} &= 2.46 \\
 Y_{new} &= \frac{2.46 \times 19.68}{7} \implies Y_{new} \approx 6.92
 \end{aligned}$$

c) Calculating INV1 input capacitance (Unknown Z)

By setting $f_3 = \hat{f}$:

$$\begin{aligned}
 f_3 &= g_3 \cdot h_3 = \hat{f} \\
 \left(\frac{4}{3}\right) \cdot \left(\frac{Z}{6.92}\right) &= 2.46 \\
 Z &= \frac{2.46 \times 6.92 \times 3}{4} \implies Z \approx 12.77
 \end{aligned}$$

d) Calculating INV2 input capacitance (Unknown W)

By setting $f_4 = \hat{f}$:

$$f_4 = g_4 \cdot h_4 = \hat{f}$$

$$\frac{W}{12.77} = 2.46$$

$$W = 2.46 \times 12.77 \implies W \approx 31.41$$

The calculated capacitance values were converted to the size parameter (scaling factor) to be used in the simulator. This was done by dividing the calculated capacitance by the input capacitance of the corresponding gate with size=1.

- NOR3 Size: $\text{Size}(X_{new}) = 19.68/7 \approx 2.81 \implies 3$
- NAND2 Size: $\text{Size}(Y_{new}) = 6.92/4 \approx 1.73 \implies 2$
- INV1 Size: $\text{Size}(Z) = 12.77/3 \approx 4.25 \implies 4$
- INV2 Size: $\text{Size}(W) = 31.41/3 \approx 10.47 \implies 10$

Based on the calculations, the circuit was implemented and simulated in the `min_delay.sp` file.

3.3 Simulation Results and Comprehensive Comparison

Table 6: Timing parameters of the optimized 5-stage path

Parameter	Measured Value
t_{pHL}	30.53 ps
t_{pLH}	39.63 ps
$t_{propagation}$	35.08 ps
t_{rise}	20.17 ps
t_{fall}	14.39 ps

Comprehensive comparison of results (3-stage vs. 5-stage path):

Table 7: Comprehensive comparison of design results

Parameter	3-Stage Circuit	5-Stage Circuit	Variations
$t_{propagation}$	33.21 ps	35.08 ps	+1.87 ps (5.6% increase)
t_{rise}	39.09 ps	20.17 ps	-18.92 ps (48.4% improvement)
t_{fall}	22.80 ps	14.39 ps	-8.41 ps (36.9% improvement)

Although the initial goal was to reduce propagation delay, the optimized 5-stage design ultimately performed slightly slower than the initial 3-stage circuit (delay increased from 33.21ps to 35.08ps). This result perfectly matches the theoretical Logical Effort predictions, which indicated higher parasitic delay for the longer path.

However, this design brought a major success in another area: a dramatic improvement in the quality of the output signal. The Rise and Fall times were reduced by 48% and 37% respectively, which translates to much sharper and stronger signal edges.